COURIER MANAGEMENT SYSTEM

(CMS)

A Salesforce Cloud-Based Project Report

Submitted by

A. Anbarasi	**24131130500112001**
S. Shenash Syed	**24131130500112015**
V. Kaviya	**24131130500112009**
B. Nandhini	**24131130500112013**

BCA Final Year

Vallalar Arts and Science College

Annamalai University

User Story

Courier and logistics companies often struggle to efficiently manage shipments and deliveries due to reliance on manual processes and disconnected systems. The absence of a centralized shipment tracking mechanism makes it difficult to monitor parcel movement in real time, resulting in delayed delivery updates and poor visibility for customers.

Delivery agent assignments are performed manually, increasing the chances of scheduling conflicts and operational delays. Furthermore, the lack of automated notifications prevents timely communication with customers, while the absence of performance analytics limits managerial insight into operational efficiency. Managing multiple branches becomes complex without an integrated platform, leading to coordination issues across locations.

Collectively, these challenges result in customer dissatisfaction, reduced productivity, operational inefficiencies, and ultimately revenue loss for the organization.

1. Introduction

1.1 Project Overview

Courier and logistics companies face challenges in efficiently managing shipments and deliveries due to manual operations and disconnected systems. These issues lead to delayed updates, poor customer visibility, and operational inefficiencies.

The Courier Management System (CMS) is a Salesforce-based cloud solution that centralizes shipment booking, tracking, delivery agent assignment, automated status updates, billing, and analytics. The system improves operational efficiency, enhances customer satisfaction, and provides real-time business insights.

1.2 Purpose

The purpose of the Courier Management System (CMS) is to automate and centralize courier and logistics operations using the Salesforce platform. It replaces manual and fragmented processes with an integrated digital solution enabling efficient shipment booking, real-time tracking, delivery agent assignment, automated notifications, billing, and performance reporting.

By providing a single, unified platform for managing shipments across multiple branches, CMS improves operational visibility, enhances customer communication, reduces delays and errors, and supports faster decision-making through analytics — ultimately increasing service efficiency, customer satisfaction, and business profitability.

2. Ideation Phase

2.1 Problem Statement

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	Branch Manager	manage shipments efficiently	tracking is manual	systems are disconnected	stressed
PS-2	Customer	track parcels	updates are delayed	no centralized system	uncertain

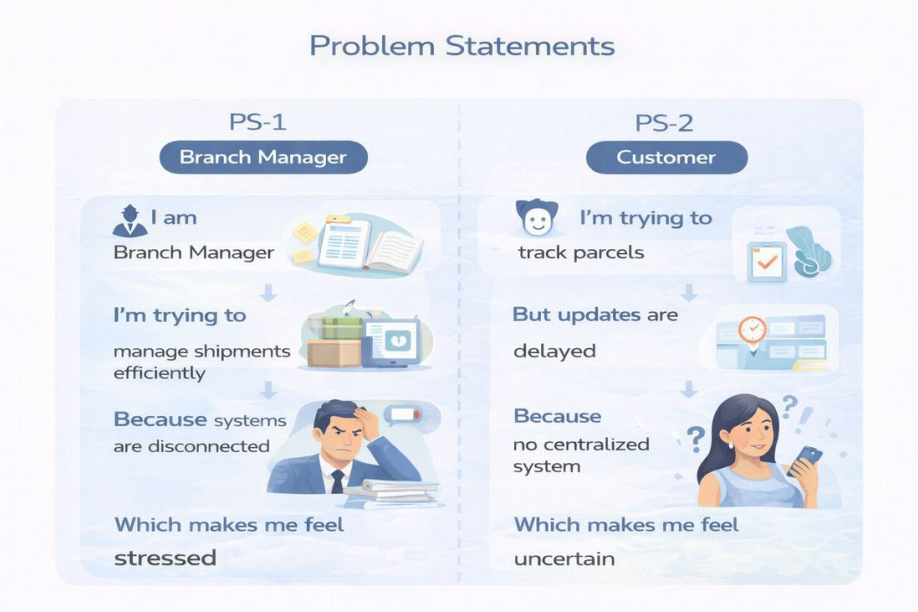


Fig 2.1 — Problem statements for the Branch Manager and Customer personas

2.2 Empathy Map Canvas

User Persona: Branch Manager

- Think & Feel: Concerned about delays & errors
- Hear: Customer complaints
- See: Registers & spreadsheets
- Say & Do: Manual follow-ups
- Pain: Delivery conflicts & missed updates
- Gain: Automated tracking & dashboards

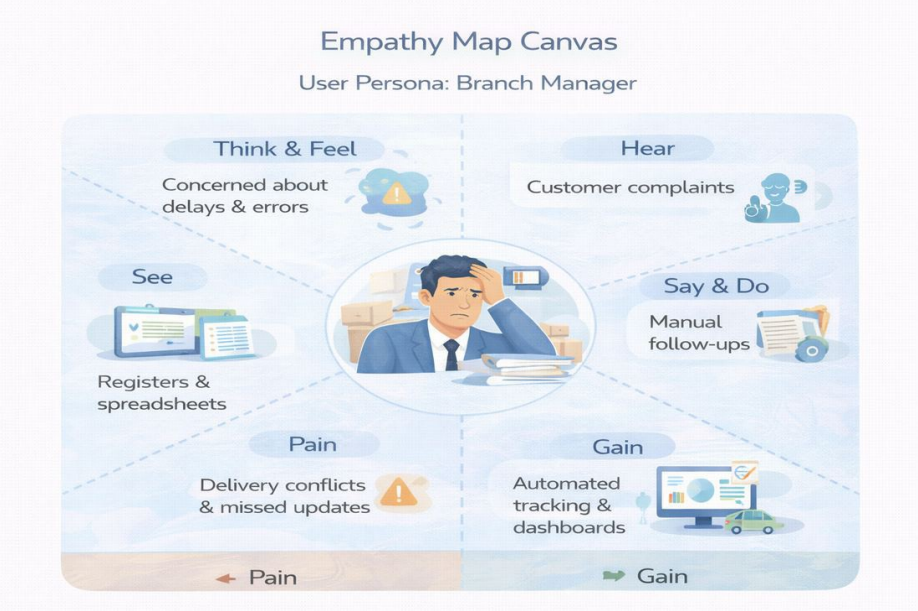


Fig 2.2 — Empathy map canvas for the Branch Manager persona

2.3 Brainstorming & Idea Prioritization

Ideas Considered

- Manual registers
- Spreadsheet tracking
- Standalone portal
- Salesforce CRM

Final Selection: Salesforce CRM

- Automation
- Scalability
- Real-time visibility

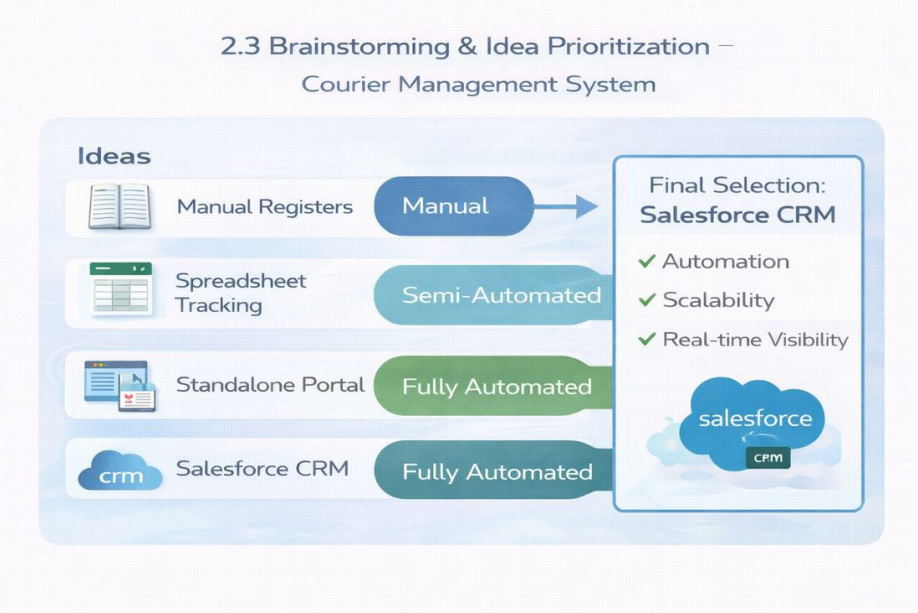


Fig 2.3 — Idea prioritization: Salesforce CRM selected over manual and semi-automated options

3. Requirement Analysis Phase

3.1 Customer Journey Map

Stages: Booking → Shipment Creation → Agent Assignment → Transit → Delivery Update → Invoice → Reporting

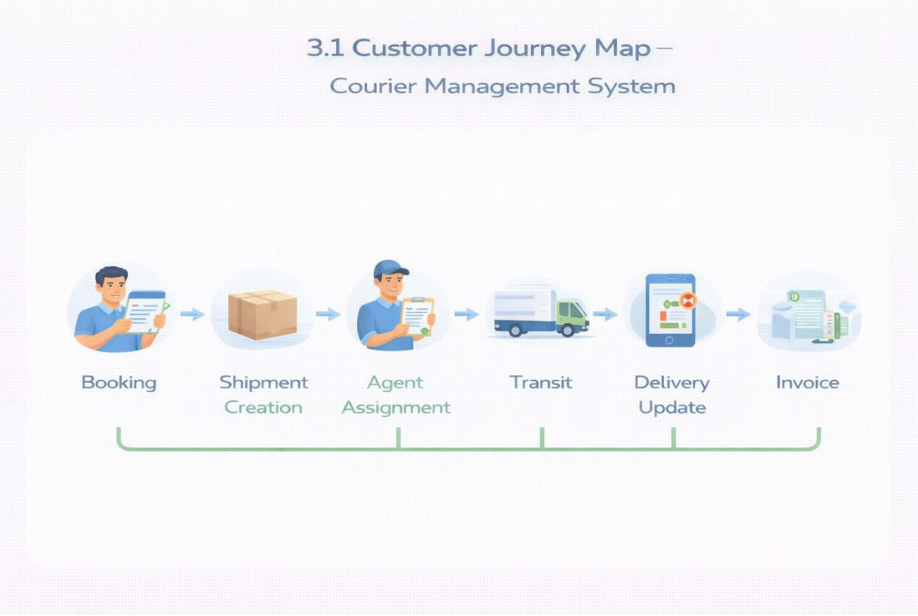


Fig 3.1 — Customer journey map across the shipment lifecycle

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	Customer management
FR-2	Shipment booking
FR-3	Delivery agent assignment
FR-4	Real-time status updates
FR-5	Invoice generation
FR-6	Reports & dashboards

Non-Functional Requirements

NFR	Description
Usability	Easy Lightning interface
Security	Role-based access control
Reliability	Error-free automation
Performance	Fast tracking updates
Availability	24/7 cloud access
Scalability	Supports growing branches

3.3 Data Flow Diagram

Flow: Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports

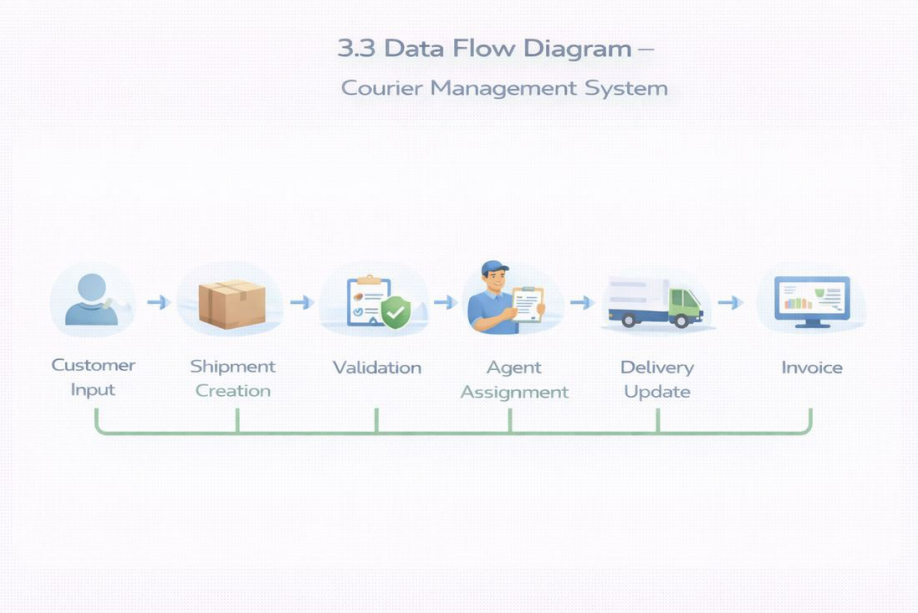


Fig 3.3 — Data flow diagram of the courier management process

3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Flows
Security	Profiles, Roles
Reporting	Reports & Dashboards

4. Project Design Phase

4.1 Problem–Solution Fit

CMS eliminates fragmented courier operations by automating shipment tracking, delivery assignment, billing, and reporting within a centralized Salesforce platform.



Fig 4.1 — Mapping each operational problem to its Salesforce CMS solution

4.2 Proposed Solution

Parameter	Description
Problem	Manual shipment management
Solution	Salesforce CMS
Innovation	Automated tracking & billing
Impact	Faster deliveries

Parameter	Description
Business Model	Courier services
Scalability	Multi-branch ready

4.3 Solution Architecture

Components

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Flows
- Reports
- Dashboards

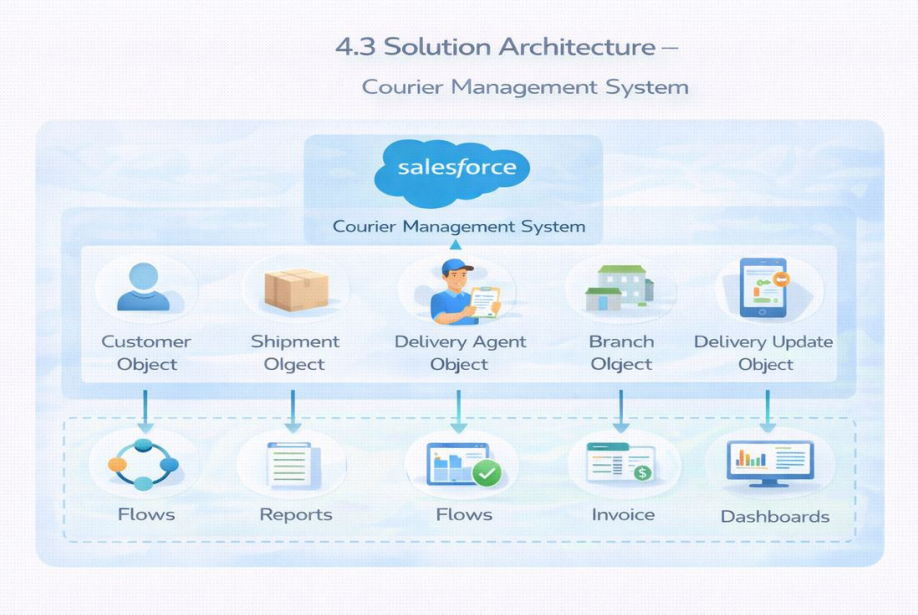


Fig 4.3 — Solution architecture connecting objects to flows, reports, and dashboards

5. Project Planning & Scheduling

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog & Sprint Schedule

Sprint	Epic	USN	User Story / Task	Pts	Priority
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the CMS.	3	High
Sprint-2	Data Modeling	USN-2	As an admin, I want to create Customer, Shipment, Delivery Agent, Branch, and Invoice objects with proper relationships so courier data is stored efficiently.	5	High
Sprint-2	Data Modeling	USN-3	As a staff member, I want tabs and a Lightning App for all courier modules so I can easily navigate and perform daily operations.	5	High
Sprint-3	Automation	USN-4	As a system, I want to validate shipment details (delivery dates, addresses, mandatory fields) so incorrect records are prevented.	3	High
Sprint-3	Automation	USN-5	As a dispatcher, I want automated flows for shipment status updates and agent assignment so deliveries are processed faster.	3	High
Sprint-4	Security	USN-6	As an administrator, I want role-based access control for managers, agents, and staff so sensitive data is secure.	5	High
Sprint-5	Reports	USN-7	As a branch manager, I want reports on delivery performance, shipment status, and revenue to monitor operations.	4	High
Sprint-6	Dashboards	USN-8	As a branch manager, I want dashboards showing delivery performance, shipment status, and revenue to monitor operations.	4	Medium

Project Tracker, Velocity & Burndown

Sprint	Total Pts	Duration	Start Date	Planned End	Completed	Release Date
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022	—	—
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022	—	—
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022	—	—

6. Project Development Phase

What You'll Learn

- Real-time Salesforce project implementation
- Designing custom objects and relationships

- Creating page layouts & record types
- Writing validation rules
- Building automation with flows
- Developing reports & dashboards
- Setting up security with profiles, permission sets, and sharing
- End-to-end application testing

Milestone 1 — Salesforce Account

A Salesforce Developer Edition account was created to build and deploy the Courier Management System. Salesforce is a cloud customer-success platform used here to model courier data, automate workflows, and generate reports.

Activity 1: Creating the Developer Account

Signed up at developer.salesforce.com with first name, last name, work email, job title (Developer), company (college name), and country (India).

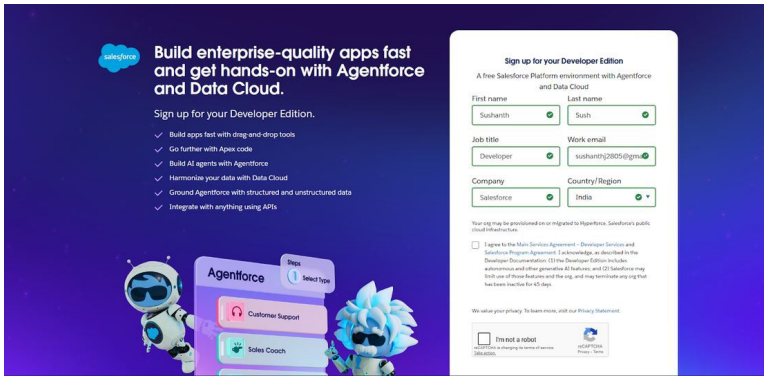


Fig 6.1 — Salesforce Developer Edition sign-up page

Activity 2: Account Activation

The verification email was opened and the account activated by setting a password and security question, redirecting to the Salesforce Setup home page.

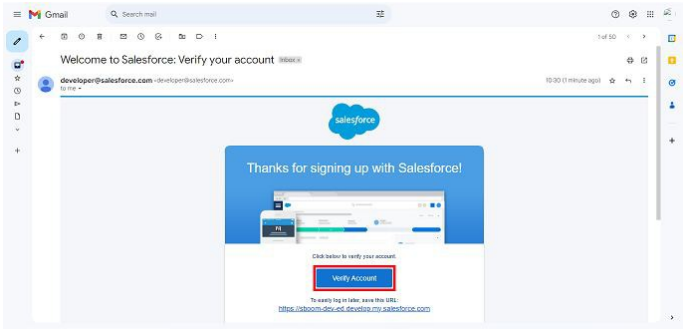


Fig 6.2 — Verification email from Salesforce

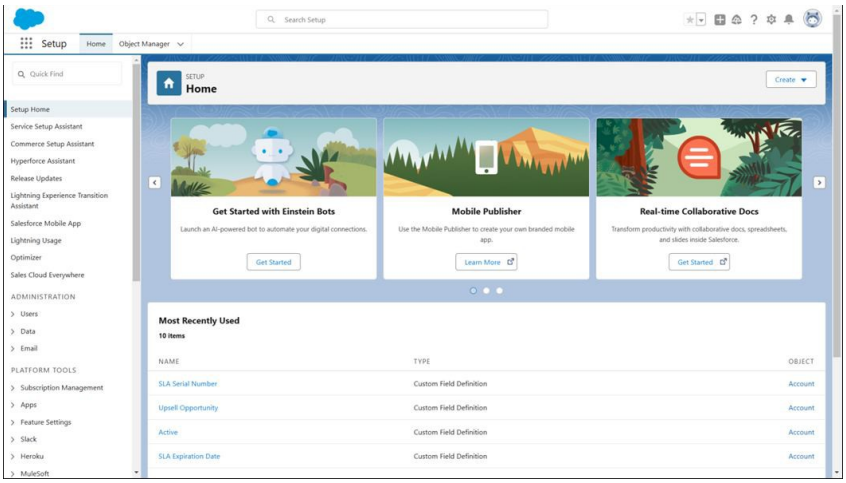


Fig 6.3 — Salesforce Setup home page after activation

Milestone 2 — Object

Salesforce objects are database tables that store organization-specific data as fields (columns) and records (rows). Six custom objects were created for CMS:

Object	Purpose
Customer__c	Stores customer personal and contact information
Shipment__c	Manages parcel details, source, destination, and tracking information
Delivery_Agent__c	Maintains delivery agent profiles and branch details
Branch__c	Represents courier branch offices
Delivery_Update__c	Stores status updates for each shipment
Invoice__c	Manages billing details for delivered shipments

Relationships

Child Object	Field	Parent Object	Type
Shipment	Customer__c	Customer	Lookup
Shipment	Assigned_Agent__c	Delivery_Agent	Lookup
Delivery_Agent	Branch__c	Branch	Lookup
Delivery_Update	Shipment__c	Shipment	Master-Detail
Invoice	Shipment__c	Shipment	Lookup

Each object was created via Setup → Object Manager → Create → Custom Object, with Allow Reports and Track Field History enabled, and Allow Search checked for Customer__c.

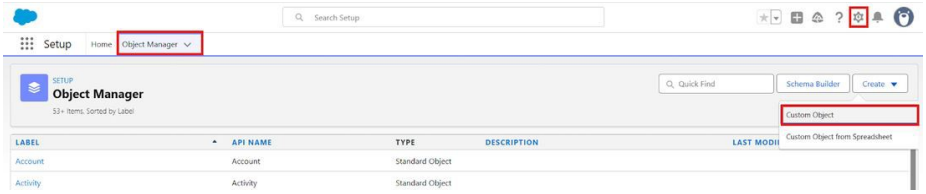


Fig 6.4 — Creating the Customer custom object via Object Manager

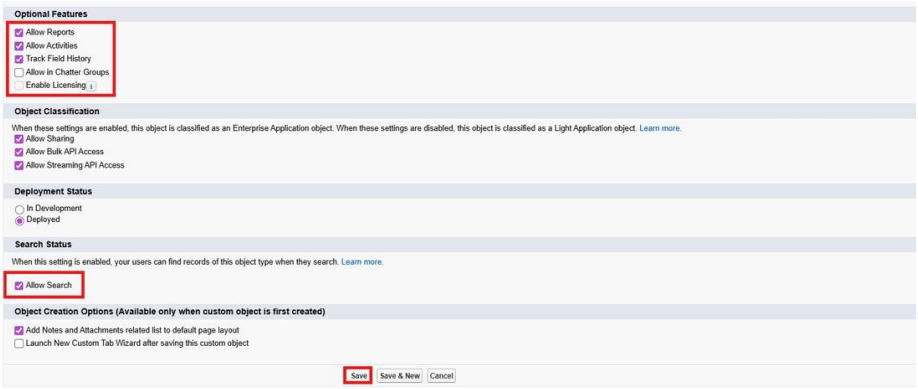


Fig 6.5 — Optional feature settings enabled for the Customer object

Milestone 3 — Tabs

Tabs let users view object data at a glance. A custom tab (Setup → Tabs → New, under Custom Object Tabs) was created for each of the six objects — Customer, Shipment, Delivery Agent, Branch, Delivery Update, and Invoice — each with its own tab style/icon and default settings.

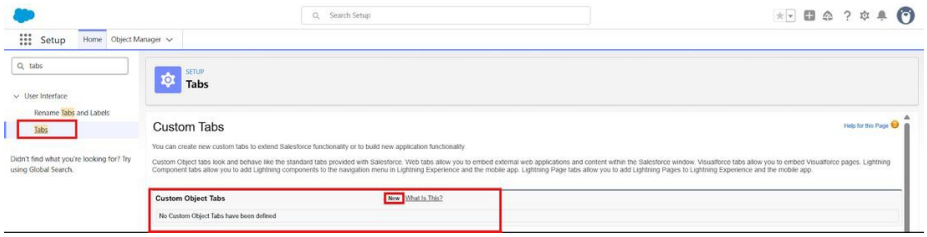


Fig 6.6 — Creating a custom tab for an object via Setup → Tabs

Milestone 4 — Fields & Relationships

Fields represent the columns of data stored on each object. The key fields created per object are summarized below.

Customer__c

Field	API Name	Type	Req.
Customer Name	Name	Text	Yes
Email	Email__c	Email	Yes
Phone	Phone__c	Phone	Yes
Address	Address__c	Text Area	Yes

Field	API Name	Type	Req.
Customer Type	Customer_Type__c	Picklist (Regular/Business)	Yes
City / State / Pincode	City__c / State__c / Pincode__c	Text	No
Is Active	Is_Active__c	Checkbox	No

Shipment__c

Field	API Name	Type	Req.
Tracking Number	Name	Auto Number (SHP-{00000})	Yes
Customer	Customer__c	Lookup(Customer)	Yes
Source / Destination Address	Source_Address__c / Destination_Address__c	Text Area	Yes
Weight	Weight__c	Number	Yes
Price	Price__c	Currency	Yes
Status	Status__c	Picklist (Booked/In Transit/Out for Delivery/Delivered/Cancelled)	Yes
Expected Date	Expected_Date__c	Date	Yes
Assigned Agent	Assigned_Agent__c	Lookup(Delivery_Agent)	No
Shipment Type	Shipment_Type__c	Picklist (Domestic/International)	Yes

Delivery_Agent__c, Branch__c, Delivery_Update__c, Invoice__c

Object	Key Fields
Delivery_Agent__c	Agent Name (Text), Email, Phone, Branch (Lookup), Status (Active/Inactive/On Leave), Joining Date
Branch__c	Branch Name (Text), Branch Code, City, State, Manager Name
Delivery_Update__c	Update Number (Auto Number UPD-{0000}), Shipment (Master-Detail), Status (In Transit/Out for Delivery/Delivered/Delayed), Update Date, Remarks
Invoice__c	Invoice Number (Auto Number INV-{0000}), Shipment (Lookup), Amount (Currency), Invoice Date, Payment Status (Paid/Unpaid/Pending)

Relationships were built as Lookup or Master-Detail fields directly on the child object (e.g. Setup → Shipment__c → Fields → New → Lookup Relationship → related object Customer__c), matching the relationship table under Milestone 2.

Milestone 5 — Validation Rules

Validation rules block a record from saving until the entered data meets the required criteria.

Object	Rule Name	Formula	Error Message
Customer__c	Email_Required	ISBLANK(Email__c)	Email is required.

Object	Rule Name	Formula	Error Message
Customer__c	Validate_Phone_Number	LEN(Phone__c) <> 10	Phone number must be 10 digits.
Customer__c	Validate_Pincode	LEN(Pincode__c) <> 6	Pincode must be 6 digits.
Shipment__c	Validate_Weight	Weight__c <= 0	Weight must be greater than zero.
Shipment__c	Validate_Price	Price__c <= 0	Price must be greater than zero.
Shipment__c	Validate_Expected_Date	Expected_Date__c < TODAY()	Expected delivery date cannot be in the past.
Delivery_Agent__c	Validate_Agent_Email	NOT(CONTAINS(Email__c,"@"))	Enter a valid email address.
Branch__c	Branch_Code_Required	ISBLANK(Branch_Code__c)	Branch Code is mandatory.
Delivery_Update__c	Remarks_Required_For_Delay	AND(ISPICKVAL(Status__c,"Delayed"), ISBLANK(Remarks__c))	Remarks are required when status is Delayed.

Milestone 6 — Page Layouts

Page layouts control how fields, sections, and related lists appear on each object's record page. Custom sections were arranged for every object:

Object	Sections Configured
Customer__c	Customer Information (Name, Type, Email, Phone, Is Active); Address Details (Address, City, State, Pincode)
Shipment__c	Shipment Details (Tracking Number, Customer, Type, Weight, Price, Status); Address Information (Source, Destination); Delivery Information (Expected Date, Assigned Agent)
Delivery_Agent__c	Agent Details (Name, Email, Phone, Branch, Status, Joining Date); Additional Information (Related Shipments, Related Delivery Updates)
Branch__c	Branch Information (Name, Code, City, State, Manager Name)
Delivery_Update__c	Update Information (Update Number, Shipment, Status, Update Date, Remarks)
Invoice__c	Invoice Details (Invoice Number, Shipment, Amount, Invoice Date, Payment Status)

Milestone 7 — Flows

A record-triggered flow automates business logic without code. The flow below keeps Shipment status in sync with the latest Delivery Update.

Scenario: Auto-Update Shipment Status When Delivery Update Is Created

- Trigger: Object = Delivery_Update__c, when a record is created, condition = none (always run), optimized for Actions and Related Records.
- Update Records element (Update_Shipment_Status): finds the Shipment__c where Id equals {!$Record.Shipment__c}.
- Field mapping: Shipment.Status__c ← {!$Record.Status__c} (copies the new status from the Delivery Update to the linked Shipment).
- Saved as Auto_Update_Shipment_Status and Activated.



Fig 6.7 — Setup → Flows, starting a new flow

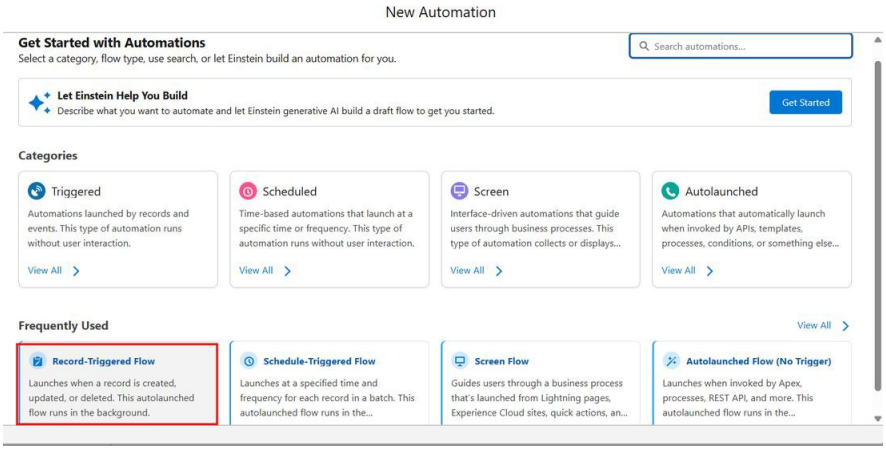


Fig 6.8 — Choosing the Record-Triggered Flow automation type

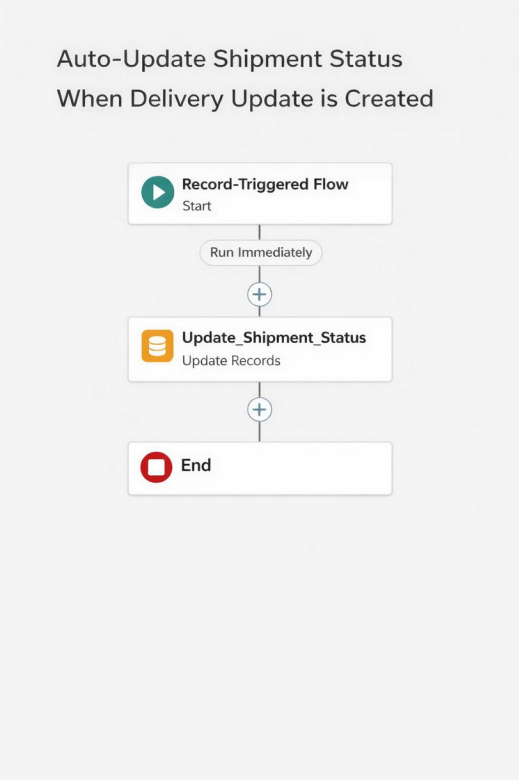


Fig 6.9 — Final flow: Delivery Update creation auto-updates Shipment status

Milestone 8 — The Lightning App

A Lightning App groups related tabs into a single working unit. The Courier Management System app was created via Setup → App Manager → New Lightning App, with the tabs Customer, Shipment, Delivery Agent, Branch, Delivery Update, Invoice, Reports, and Dashboards, and access granted to the System Administrator profile.

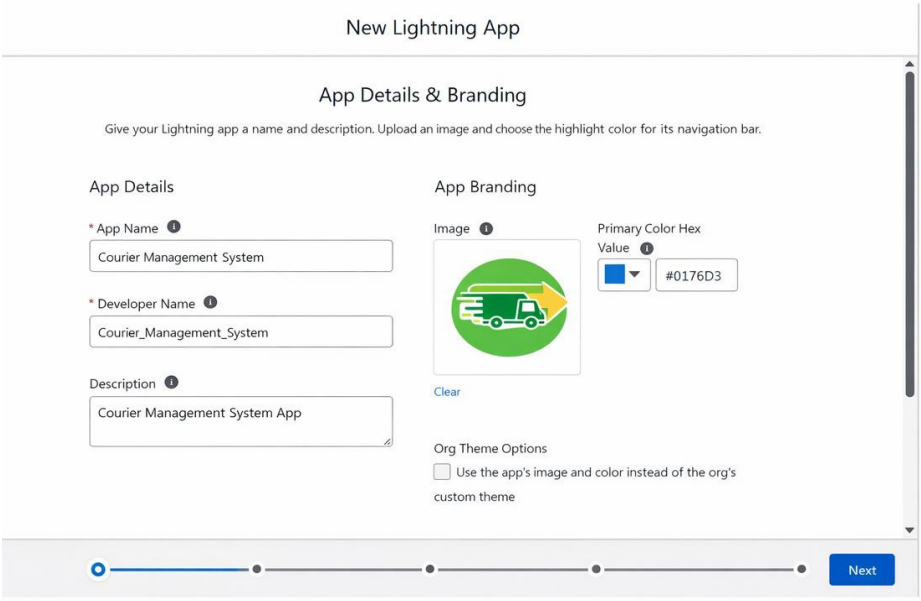


Fig 6.10 — Naming and branding the Courier Management System Lightning App

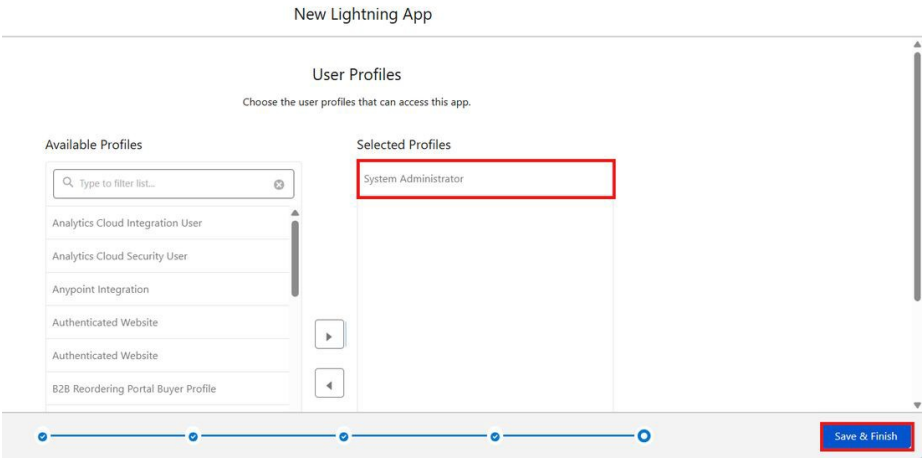


Fig 6.11 — Assigning user profiles to the new app

Milestone 9 — Profile

A profile controls what a user can see and do — object, field, tab, and app permissions, page layouts, and login restrictions. Four profiles were created for CMS:

Profile	Purpose
Courier Admin	Full system control (cloned from System Administrator)
Branch Manager	Manage shipments and agents
Delivery Agent	Update delivery status
Customer Support	Handle customers and tracking

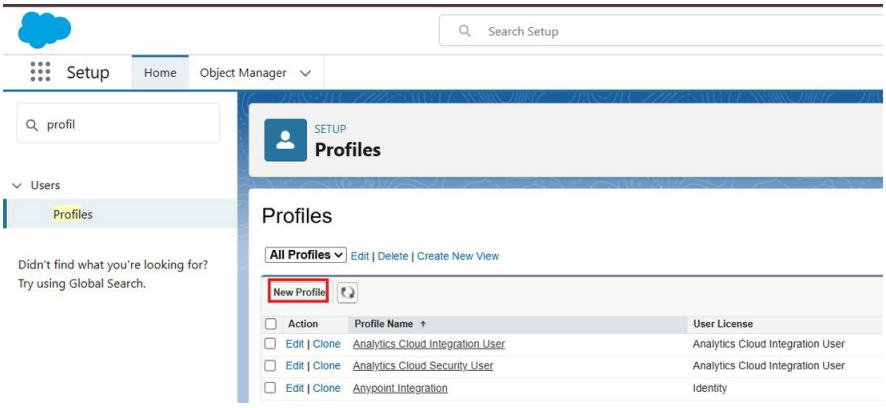


Fig 6.12 — Setup → Profiles list

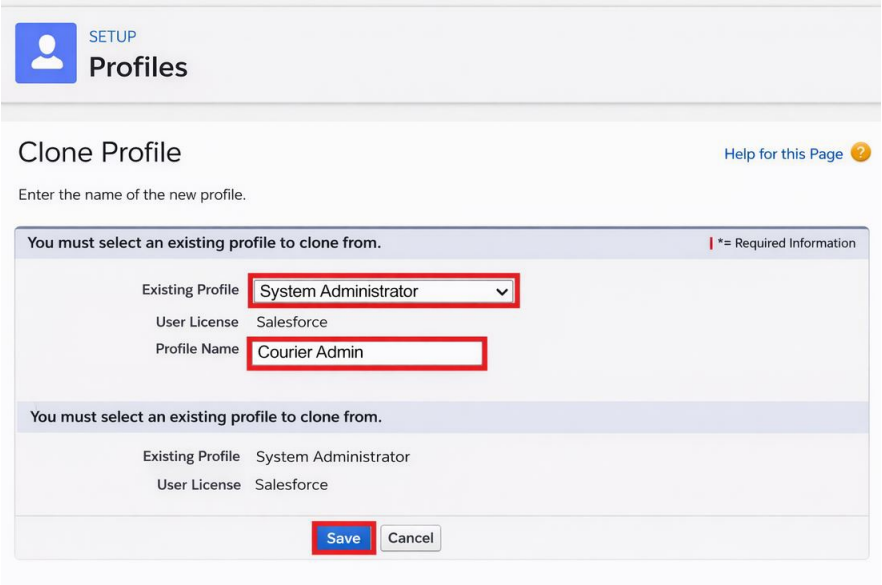


Fig 6.13 — Cloning System Administrator to create the Courier Admin profile

Object Permissions — Branch Manager

Object	Permissions
Customer__c	Create, Read, Edit
Shipment__c	Create, Read, Edit
Delivery_Agent__c	Read, Edit
Branch__c	Read, Edit
Delivery_Update__c	Create, Read, Edit
Invoice__c	Read

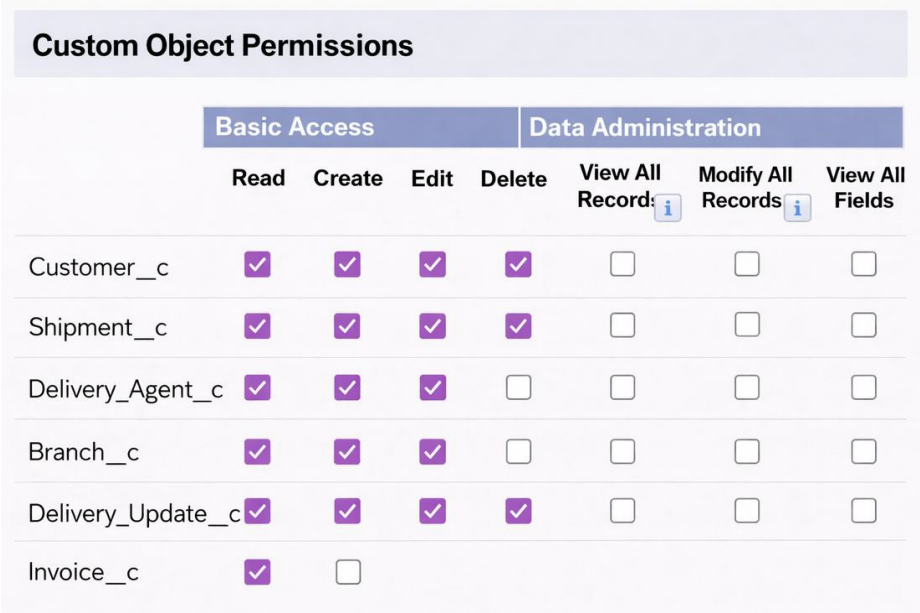


Fig 6.14 — Custom object permissions configured for Branch Manager

Session timeout was set to 2 hours of inactivity and minimum password length to 8 characters for Branch Manager. The Delivery Agent profile was scoped to Read/Edit on Shipment and Delivery Update, Read-only on Customer and Invoice, with no Branch-editing or delete rights. Customer Support was scoped to Read/Edit on Customer and Shipment, Read-only on Delivery Update.

Milestone 10 — Role

Roles define the reporting hierarchy and control which users can see each other's data — higher roles see more. A four-level role hierarchy was built under the CEO role:

- Courier Admin — highest level, access to all data
- Branch Manager (under Courier Admin) — manages branch-level operations
- Delivery Agent (under Branch Manager) — access only to assigned shipments
- Customer Support (under Branch Manager) — limited access to customer and shipment tracking

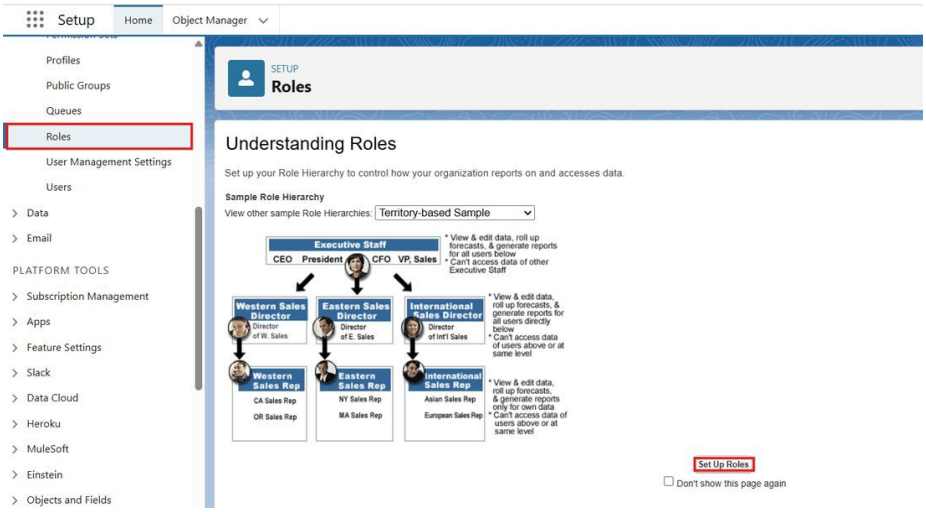


Fig 6.15 — Setup → Roles, understanding the role hierarchy

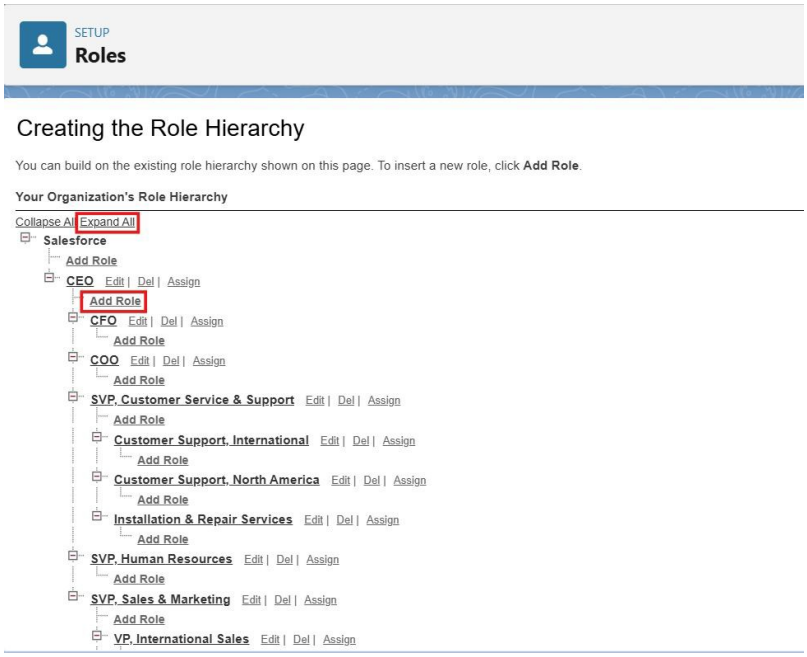


Fig 6.16 — Building the Courier Admin → Branch Manager role hierarchy

Milestone 11 — User

Every Salesforce user needs a unique username, a valid email, and an assigned role and profile. One user was created per role — Courier Admin, Branch Manager, Delivery Agent, and Customer Support — each linked to its matching profile and given the Salesforce (or Salesforce Platform) license.

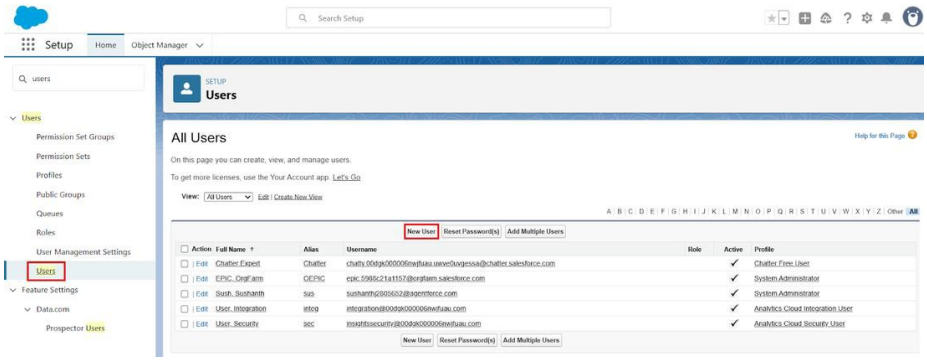


Fig 6.17 — Setup → Users, the list of created CMS users

Milestone 15 — Reports & Dashboards

Custom Report Types

- Shipment Tracking Report — Primary Object: Shipment__c
- Customer Summary Report — Primary Object: Customer__c
- Agent Performance Report — Primary Object: Delivery_Agent__c

Reports Built

Report	Key Columns / Grouping
Shipment Status Summary	Tracking Number, Customer, Status, Expected Date, Assigned Agent — grouped by Status, summarized on Price
Agent Performance Report	Agent Name, Branch, Number of Shipments, Status — grouped by Delivery Agent
Revenue by Branch Report	Tracking Number, Customer, Price, Invoice Status — grouped by Branch and Month

Courier Management Dashboard

- Shipment Status Overview — Donut Chart (from Shipment Status Summary)
- Agent Performance — Bar Chart (from Agent Performance Report)
- Revenue by Branch — Column Chart (from Revenue by Branch Report)

The dashboard was shared with the Courier Admin, Branch Managers, and Customer Support team.



Fig 6.18 — Final reports and dashboard layout for the Courier Management System

7. Functional & Performance Testing

During testing, screenshots were captured from Salesforce to verify record creation, validation rule enforcement, delivery status updates, flow automation, invoice generation, and dashboard visualization. All scenarios were validated to ensure correct performance and reliability.

8. Advantages & Disadvantages

Advantages	Disadvantages
Centralized data	Salesforce dependency

Advantages	Disadvantages
Automated workflows	Requires initial configuration effort
Real-time reporting	

9. Conclusion

The Courier Management System successfully digitizes courier operations using Salesforce automation, improving efficiency, accuracy, and customer satisfaction.

10. Future Scope

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimization

11. Appendix

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org